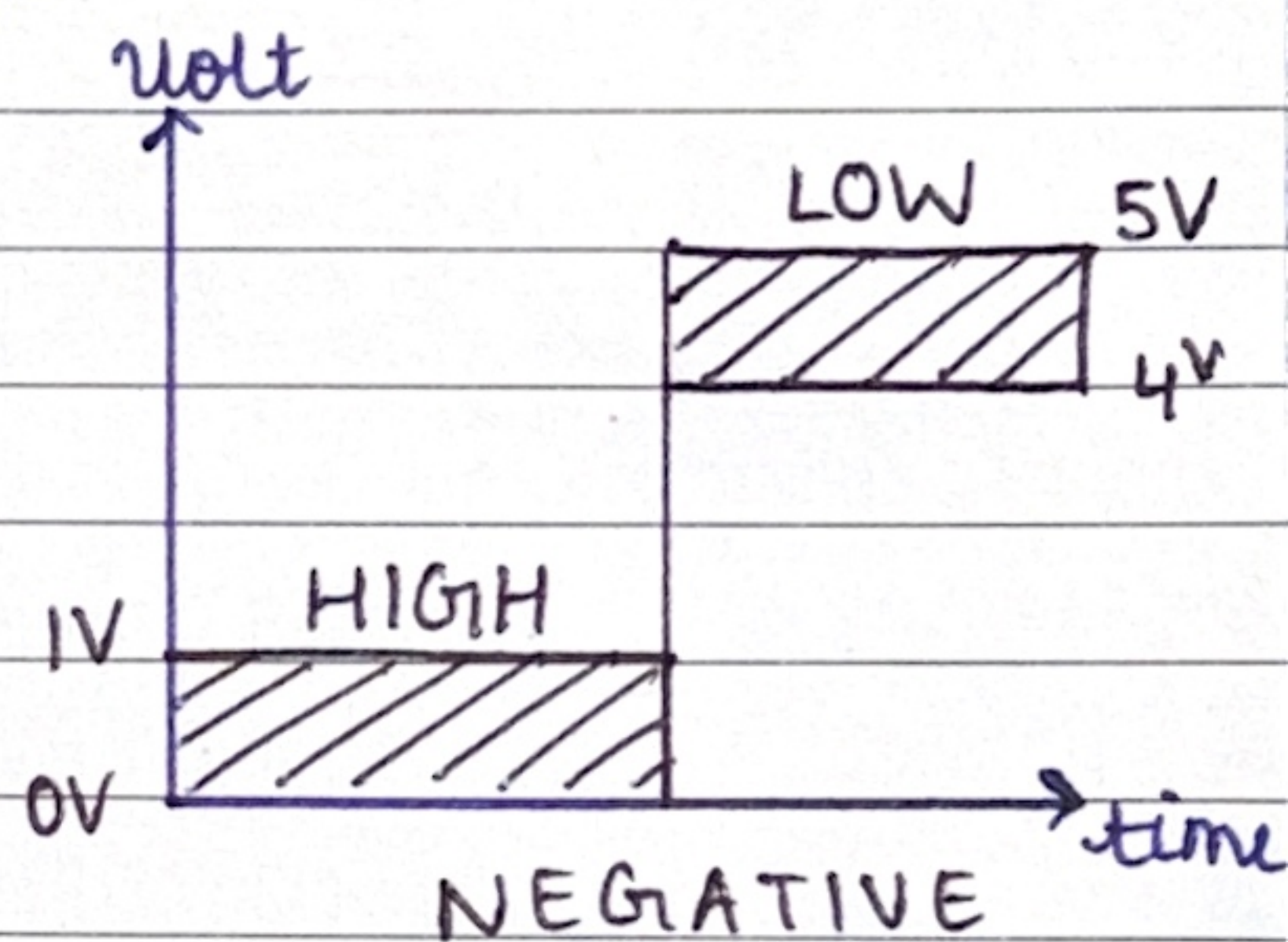
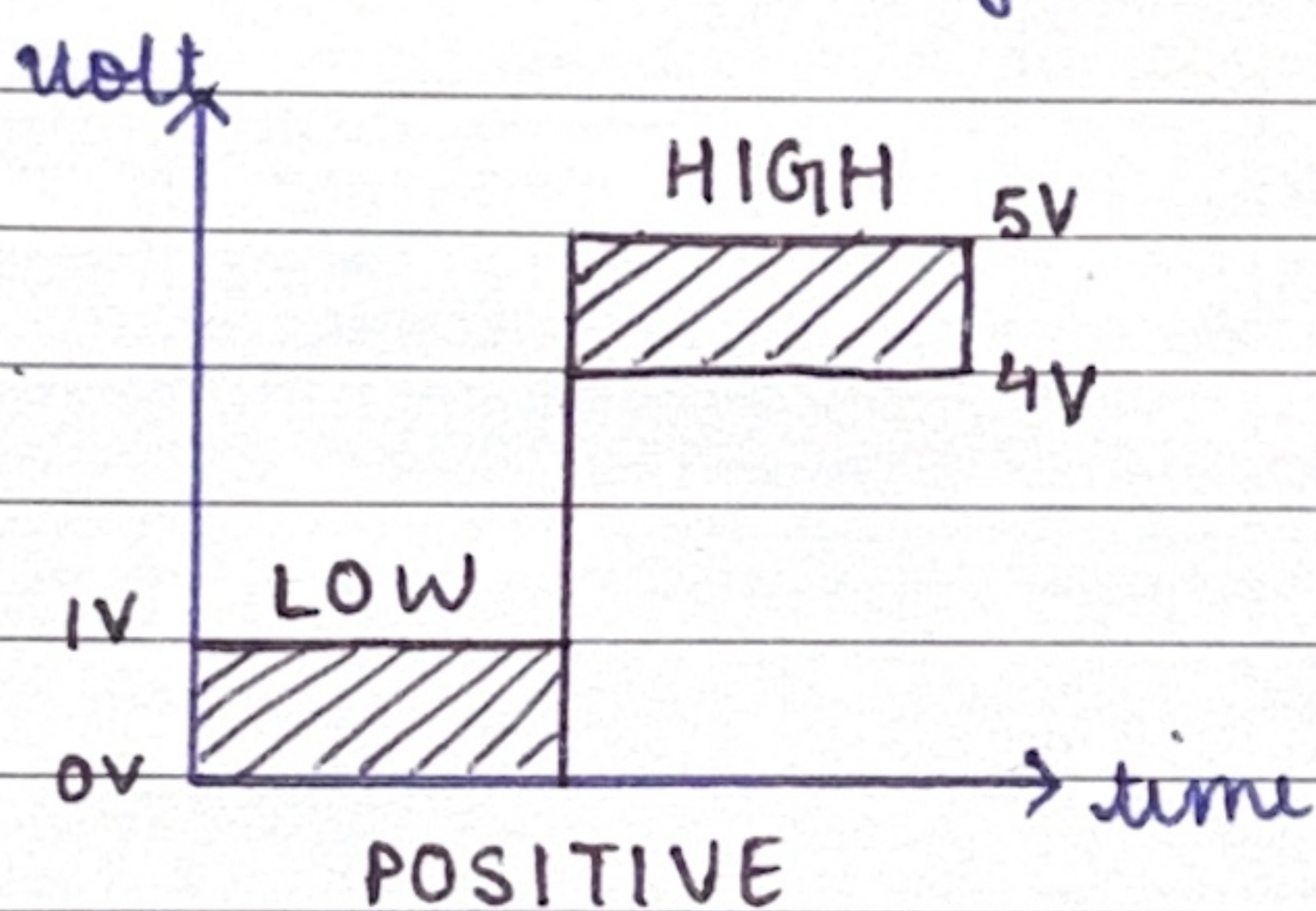


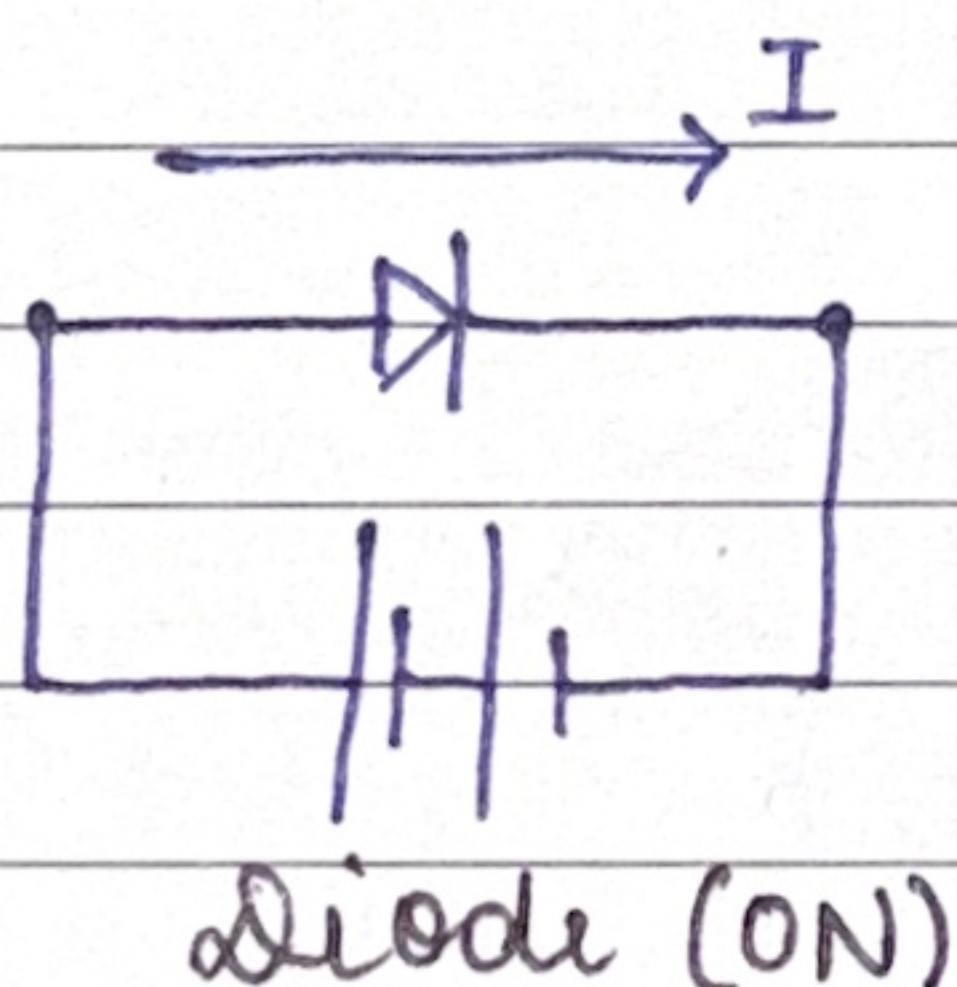
→ Digital Logic Families

- A group of compatible IC's with same logic levels and supply voltages fabricated for performing various logical functions is referred as logic family.
- There are two types of logic levels
 1. Positive Logic: Logic is high at high voltage and low on low voltage
 2. Negative Logic: Logic is high at low voltage and low at high voltage.

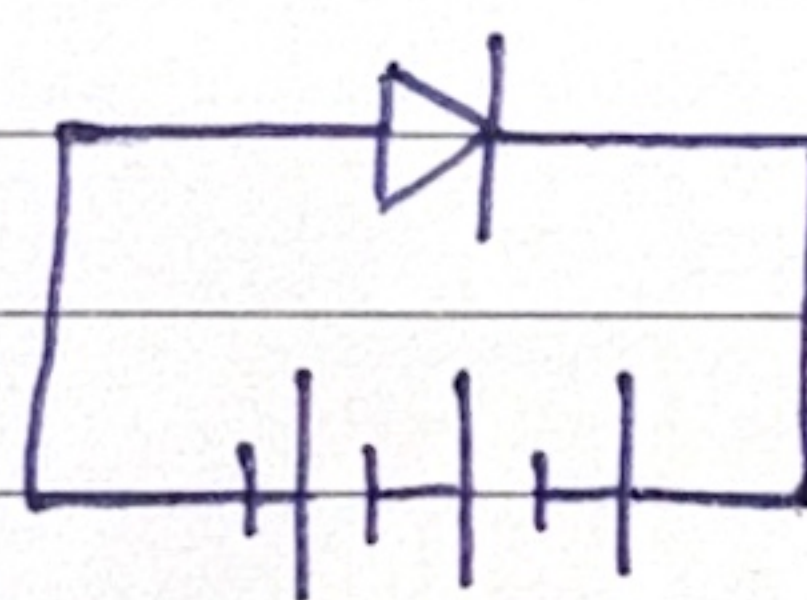


→ Semiconductor Devices

- There are two types of semiconductor devices
1. Bipolar: It has two charge carriers i.e. electrons & holes
 2. Unipolar: It has only one charge carrier i.e. either electron or holes



Diode (ON)
Forward Bias

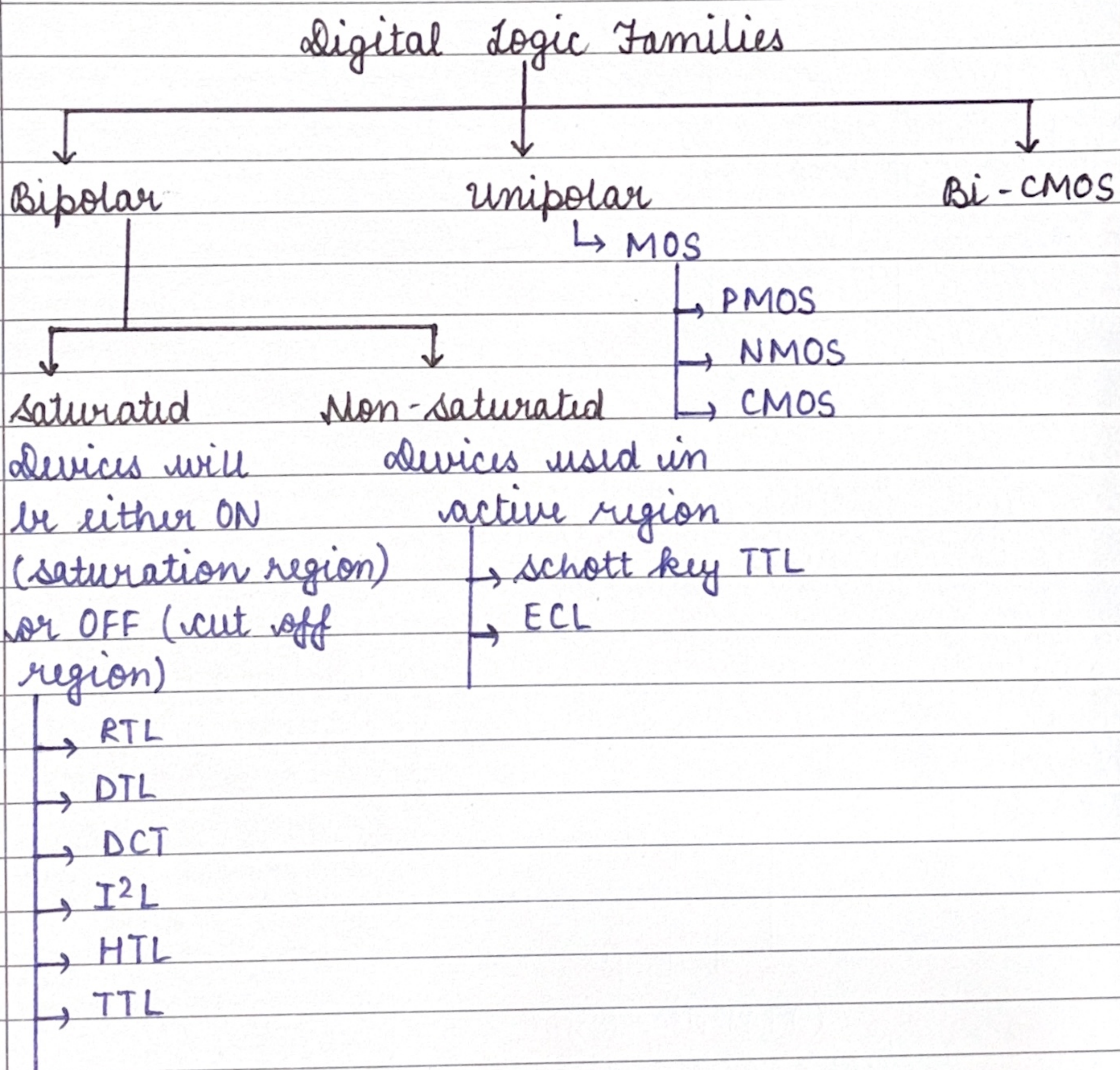


Diode (OFF)
Reverse Bias

→ No I flows in this case

→ Classification of Digital Logic Families

1. Bipolar logic families
 - Uses electrons & holes as charge carriers
2. Unipolar logic families
 - Uses either e^- or holes as charge carrier
3. Bi-CMOS logic families
 - It uses MOS that's bipolar
 - Input & logical operations performed by CMOS whereas output performed by MOS



→ Merits of digital IC

- Reduces overall size of system
- Reduces overall cost of system
- Improves reliability of system by removing inter connection of wiring
- Reduces power consumption of system

→ Demerits of digital IC

- It can't handle large power
- Electrical devices like resistor, transistor, inductor & large capacitors can't be implemented with great precision on chip
- Only suitable for low power applications

→ Characteristics of digital IC

1. Propagation Delay : It justifies speed of the operation of digital IC and explains the amount of delay happened when signal is transitioning from input to output
2. Threshold voltage : It explains the amount of voltage at which transition occurs from logic low to logic high
3. Power dissipation : It justifies the amount of power dissipated on IC in form of heat
4. Figure of merit : It's the product of the propagation delay & power dissipation
5. Fan out & Fan In : Fan out explains how many similar gates can be driven by

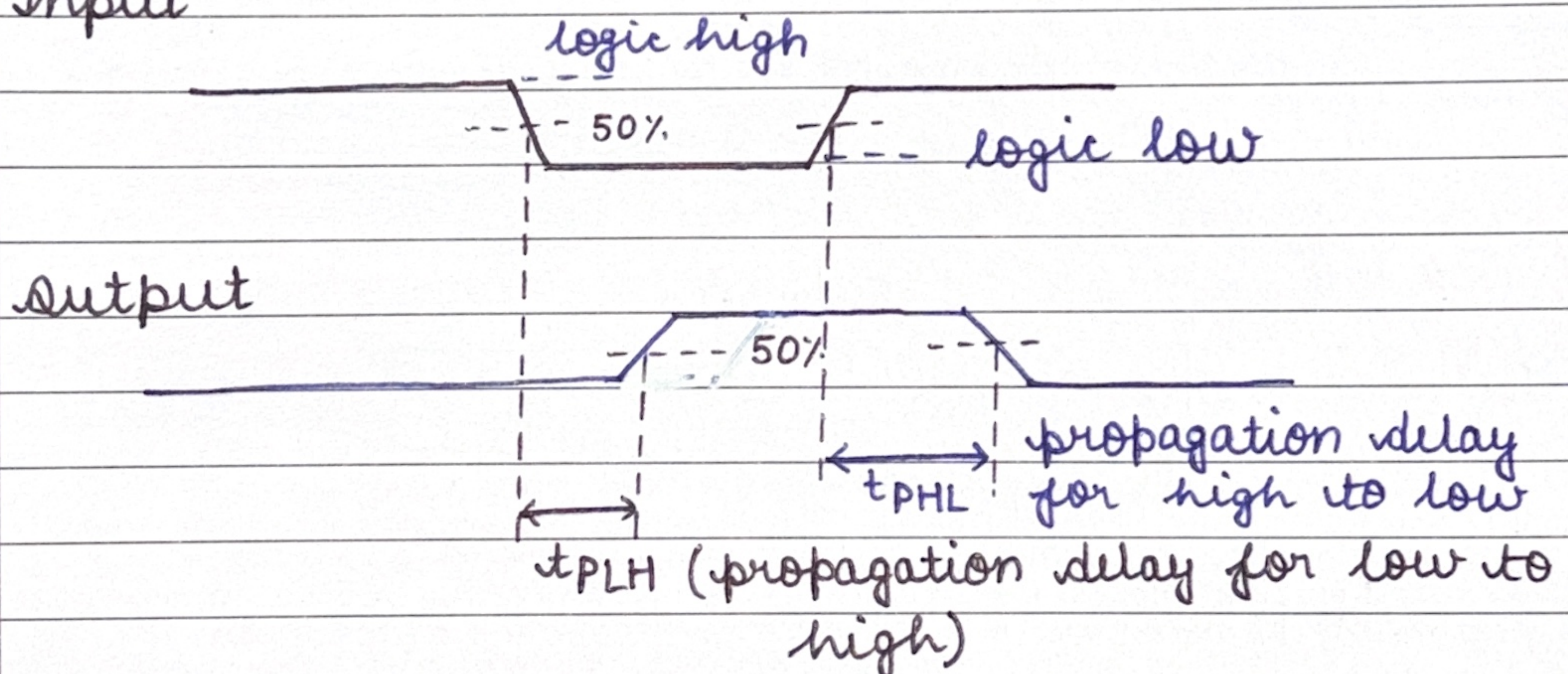
given logic gate. Fan in explains no. of inputs that can be handled at a given logical gate.

6. Noise Immunity

7. Operating Time



Input



$$t_p = \frac{t_{PLH} + t_{PHL}}{2} \rightarrow \text{Propagation Delay}$$

$$P_d = V_{cc} \times I_{cc} (\text{avg}) \quad \text{where } I_{cc} = \frac{I_{cch} + I_{ccl}}{2}$$

$\hookrightarrow V_{cc}$ in TTL & V_{DD} in CMOS

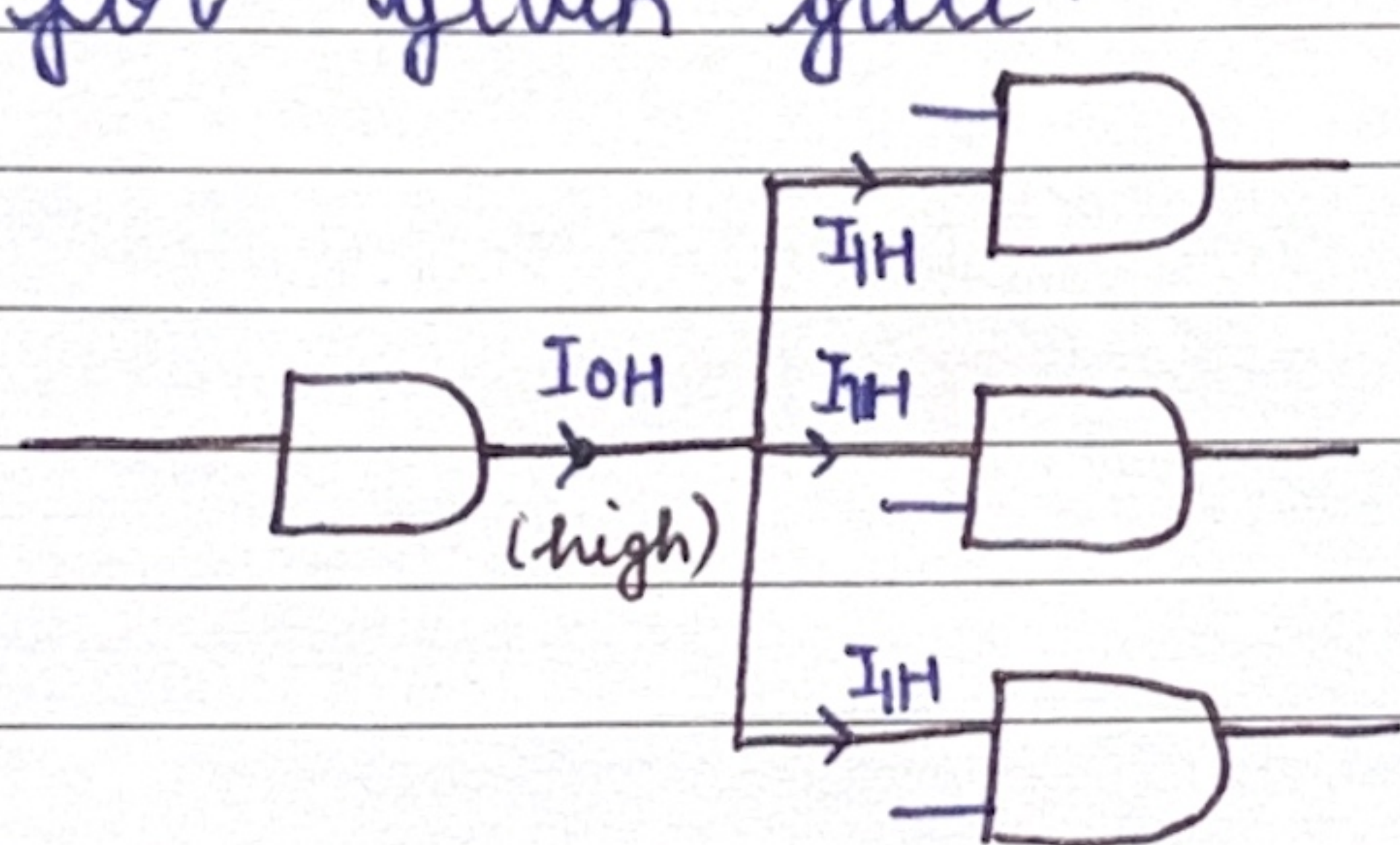
$I_{cch} \rightarrow I_{cc}$ at logic high

$I_{ccl} \rightarrow I_{cc}$ at logic low

$$\Rightarrow P_d = V_{cc} \cdot \left[\frac{I_{cch} + I_{ccl}}{2} \right] \rightarrow \text{Power dissipation}$$

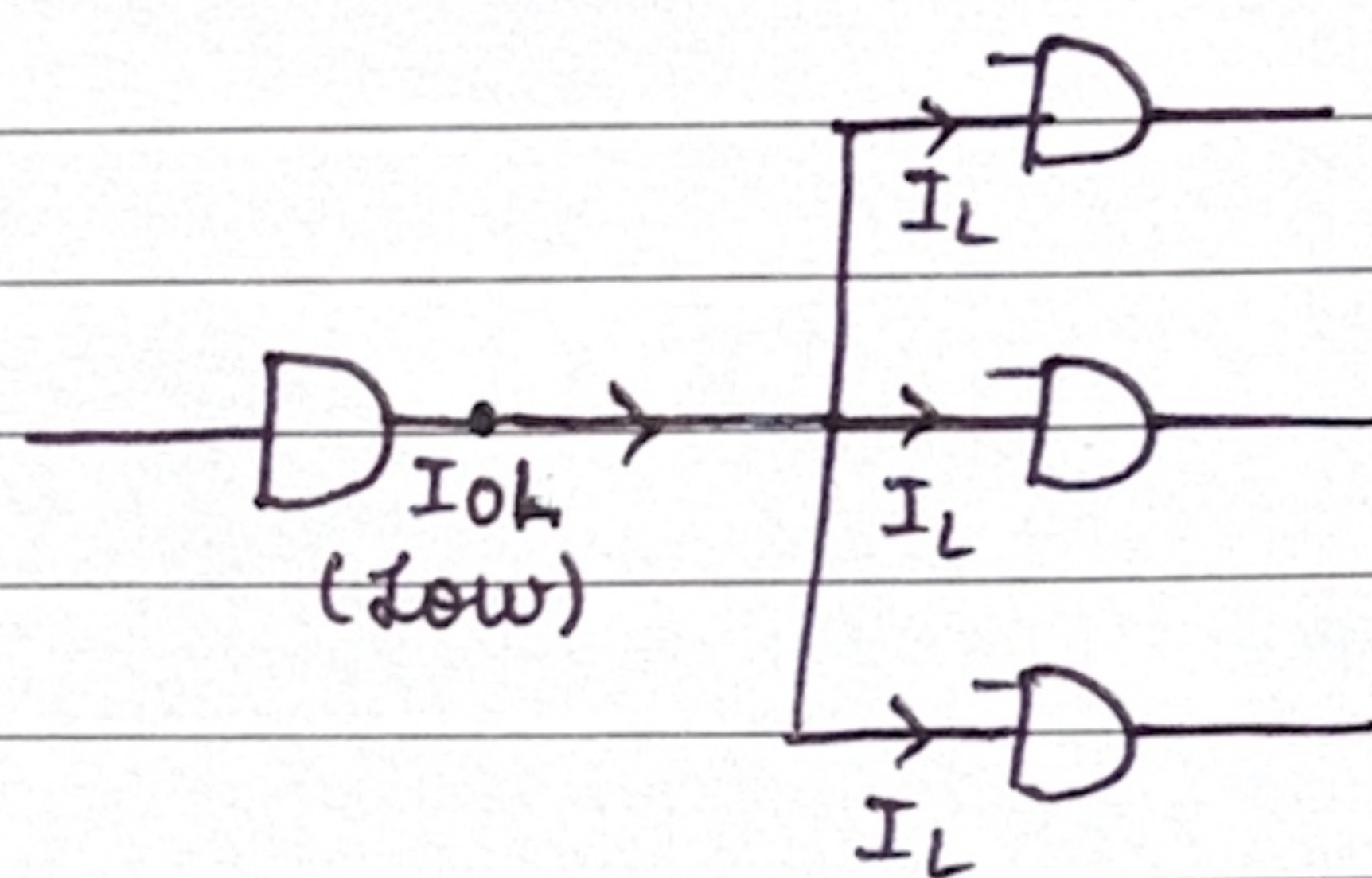
Fig of merit = $P_d \times t_p$ $\rightarrow$ good IC always have low figure of merit

- Fan out should be as high as possible so that we can drive large no of gates by single gate. It defines current supply for given gate.



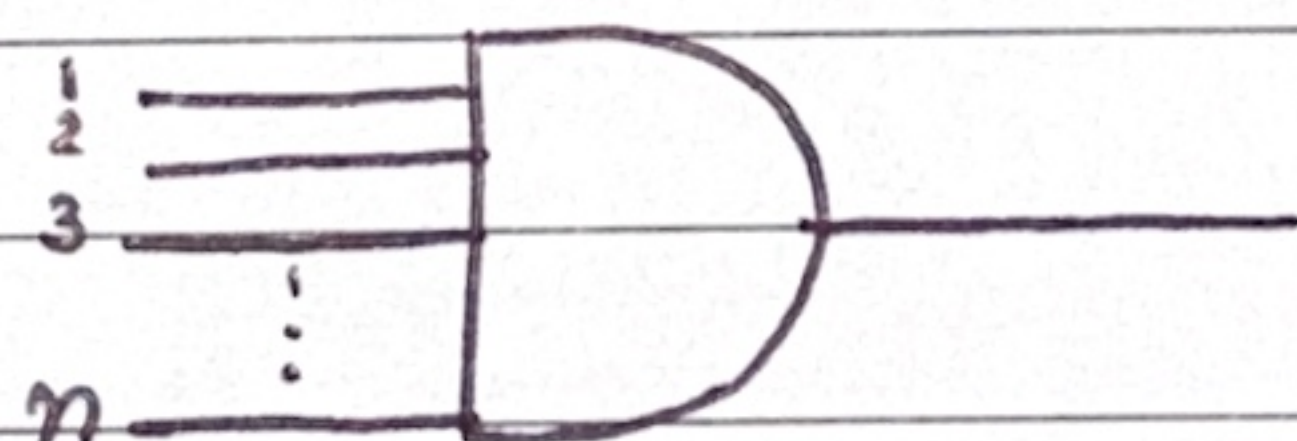
High state fan out

$$= \frac{I_{OH}(\text{max})}{I_{IH}}$$



Low state fan out

$$= \frac{I_{OL}(\text{min})}{I_{IL}}$$

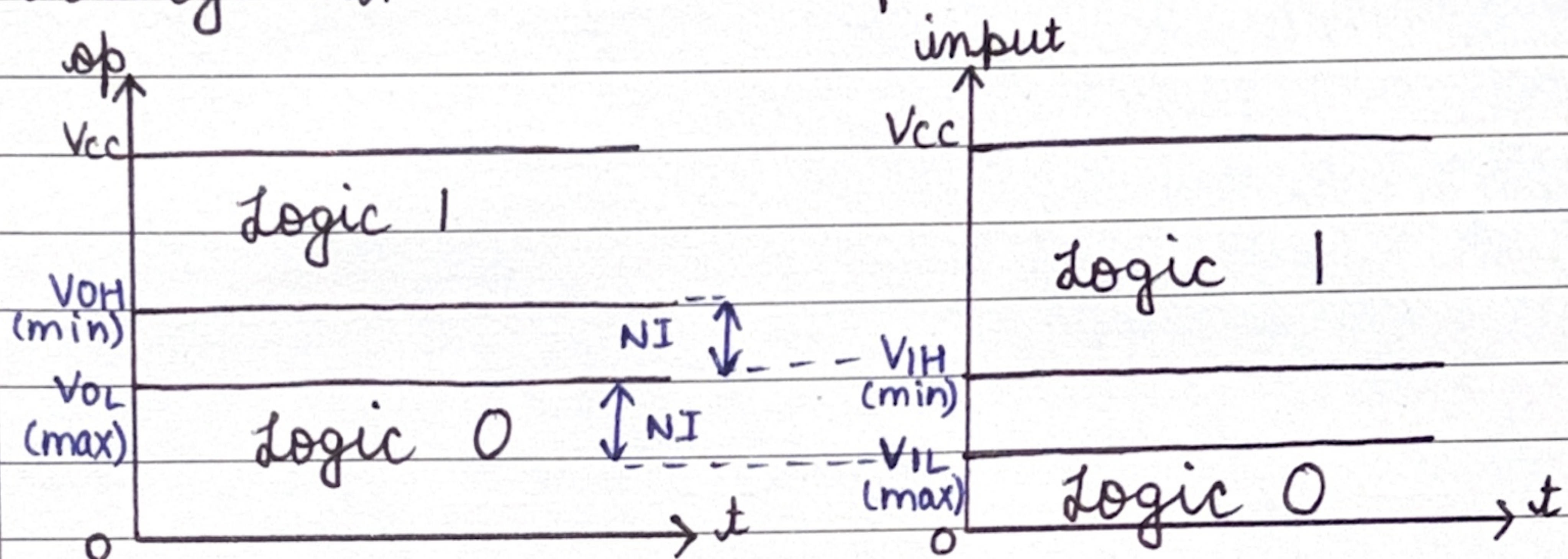


Fan In = n

7. Operating Temperature

- For commercial applications ($0-70^{\circ}\text{C}$)
- For industrial applications ($0-85^{\circ}\text{C}$)
- For military applications ($-55-125^{\circ}\text{C}$)

→ Voltage and current parameters



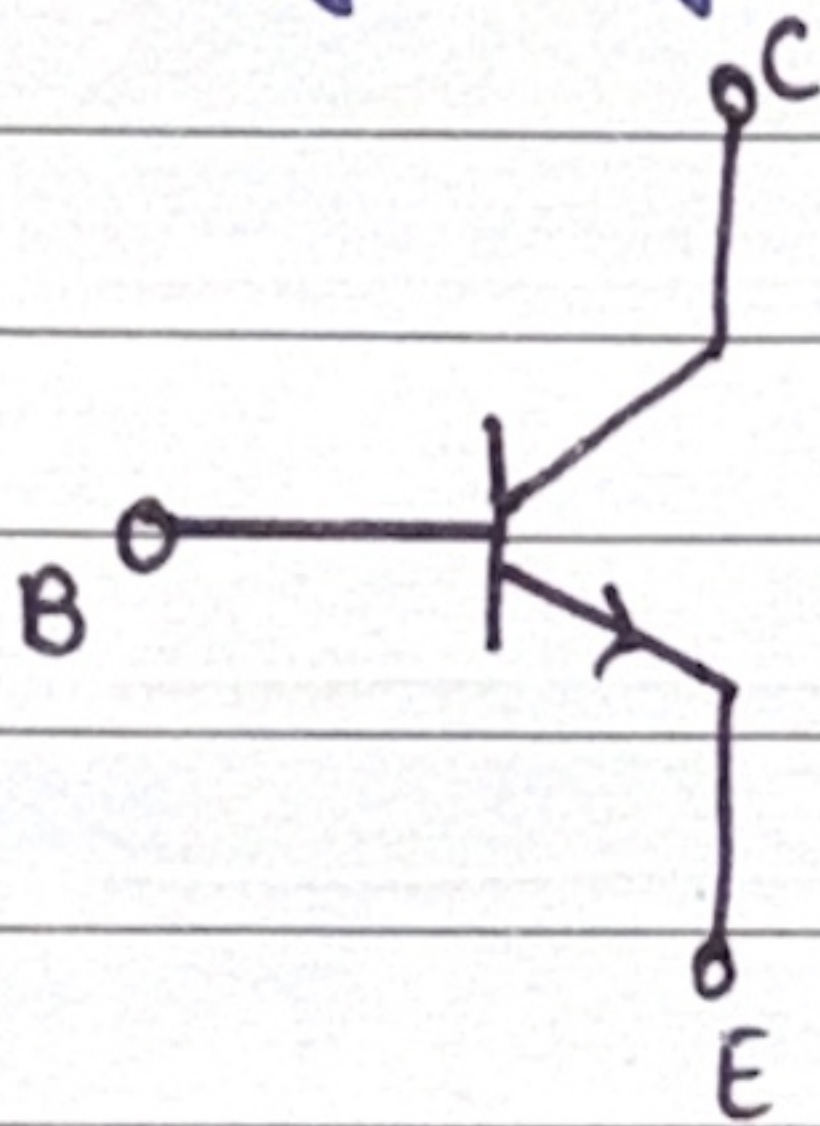
- $V_{OH(min)}$ is the minimum output voltage for logic 1
- $V_{OL(max)}$ is the maximum output voltage for logic 0
- $V_{IL(max)}$ is maximum input voltage for logic 0
- $V_{IH(min)}$ is minimum input voltage for logic 1
- I_{OH} → high level output current
- I_{OL} → low level output current
- I_{IH} → high level input current
- I_{IL} → low level input current

$V_{OH(min)} - V_{IH(min)}$ → Noise immunity for logic high
 $V_{IL(max)} - V_{OL(max)}$ → Noise immunity for logic low

→ Resistor - Transistor Logic (RTL)

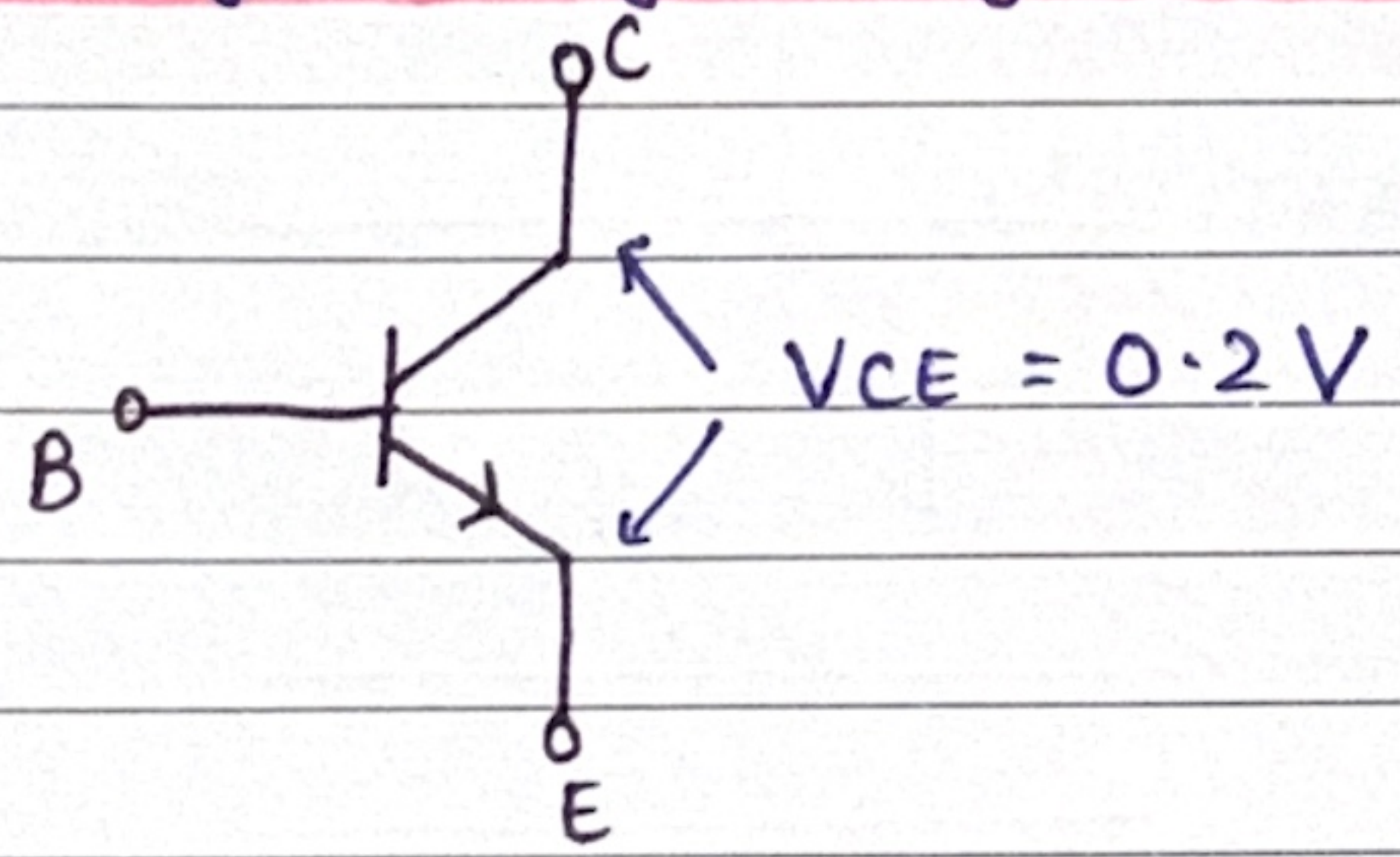
- This logic family includes resistors and transistors in its IC. So it's called as RTL family
- Here we use transistor in saturated &

cut off region, so speed of RTL family is low



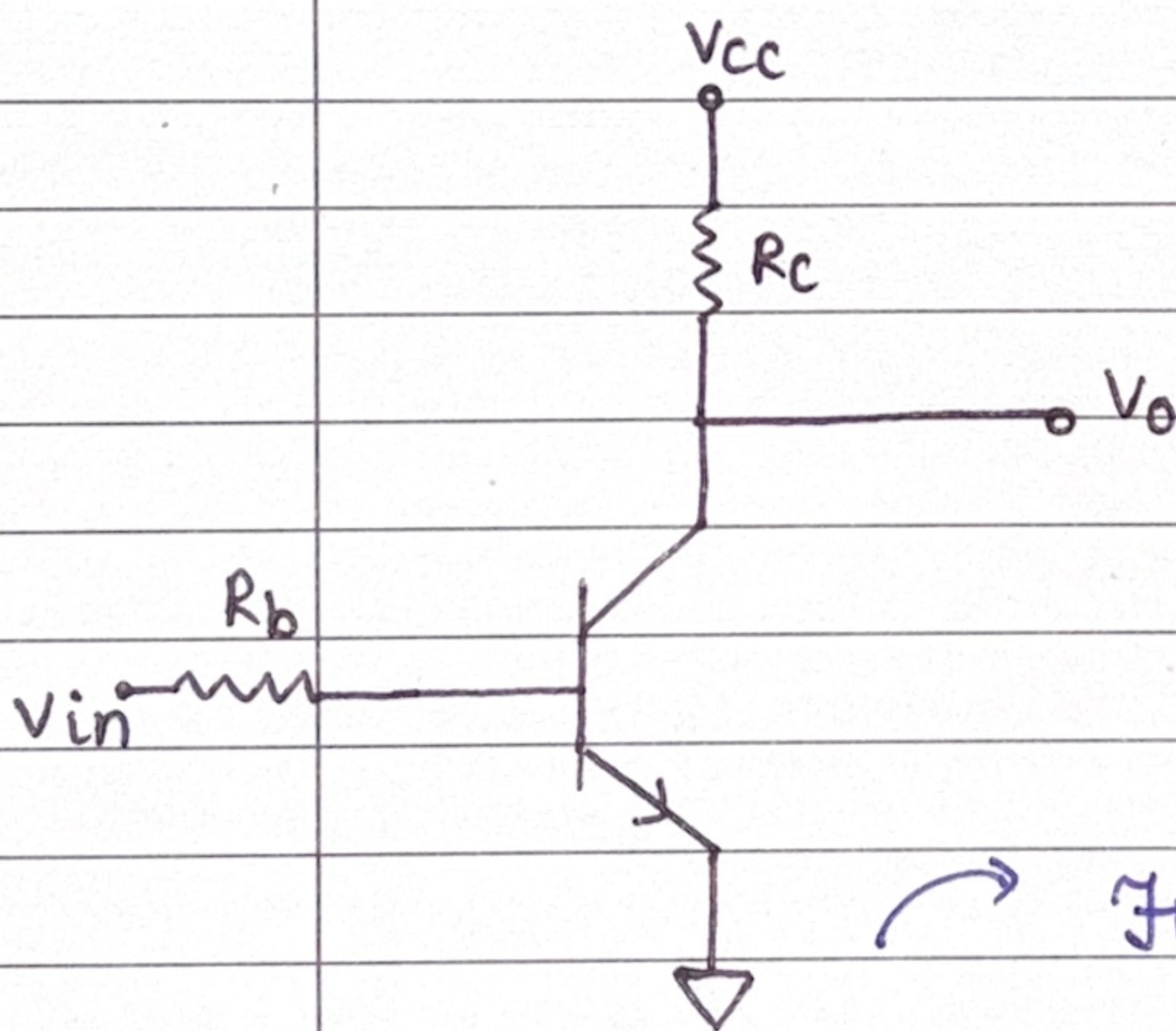
$$B = 0$$

cut off region
C to E terminal act
like open circuit



$$B = V_{CC}$$

saturation region
current flow from
C to E



Input	Output	Region
Low	High	Cut off
High	Low	Saturation

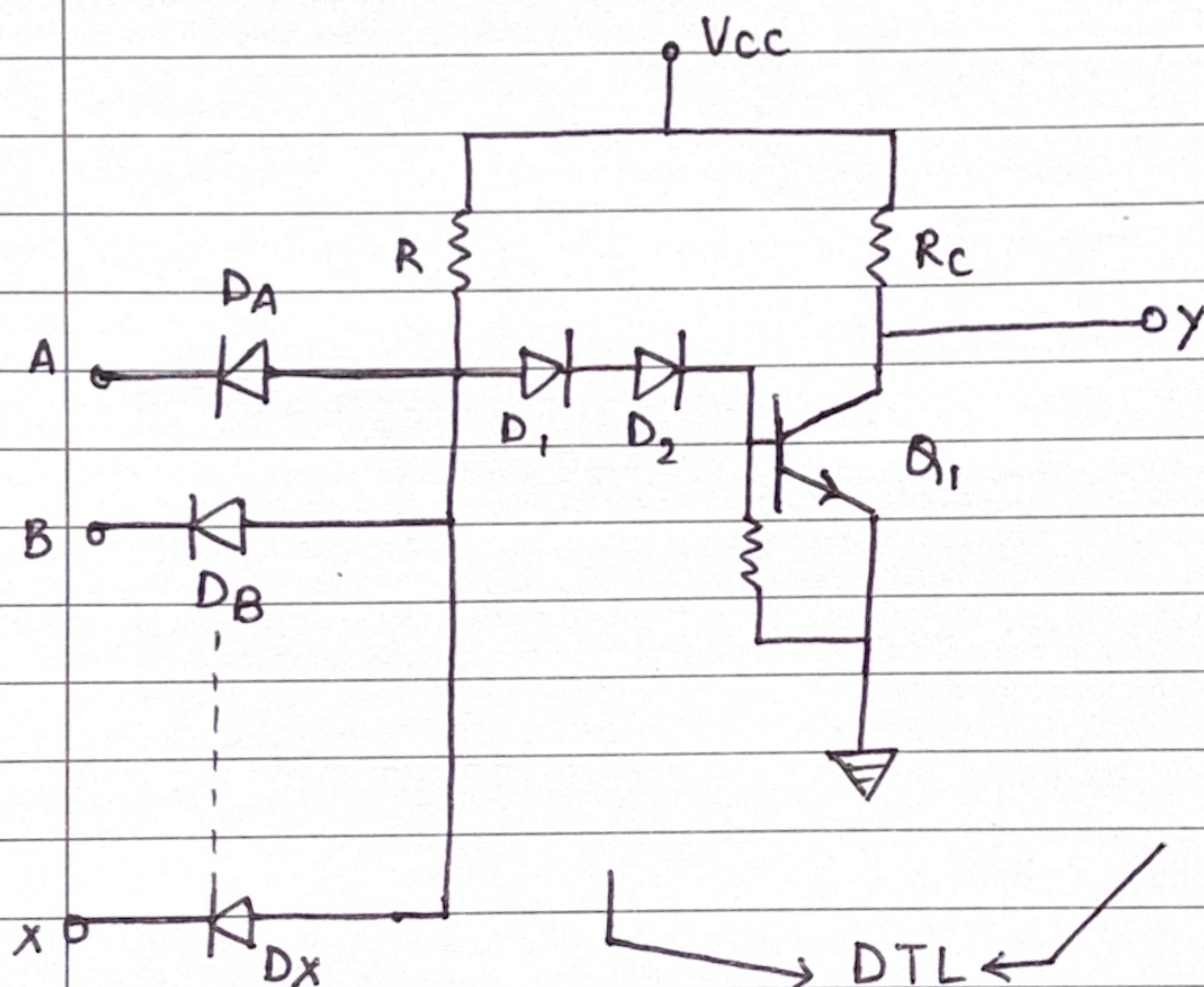
Functions as NOT gate

→ Disadvantages of RTL

- Low noise margin
- Low fan out
- Low speed
- High power dissipation due to presence of resistors

→ Diode Transistor Logic (DTL)

- RTL has low noise margin, low fan out, slow speed & higher power dissipation, so we don't use RTL in recent IC's
- DTL has improved noise margin & fan out compared to RTL



A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

DTL circuit for NAND gate

→ Disadvantages of DTL

- Lower speed
- Power dissipation is high due to use of resistors

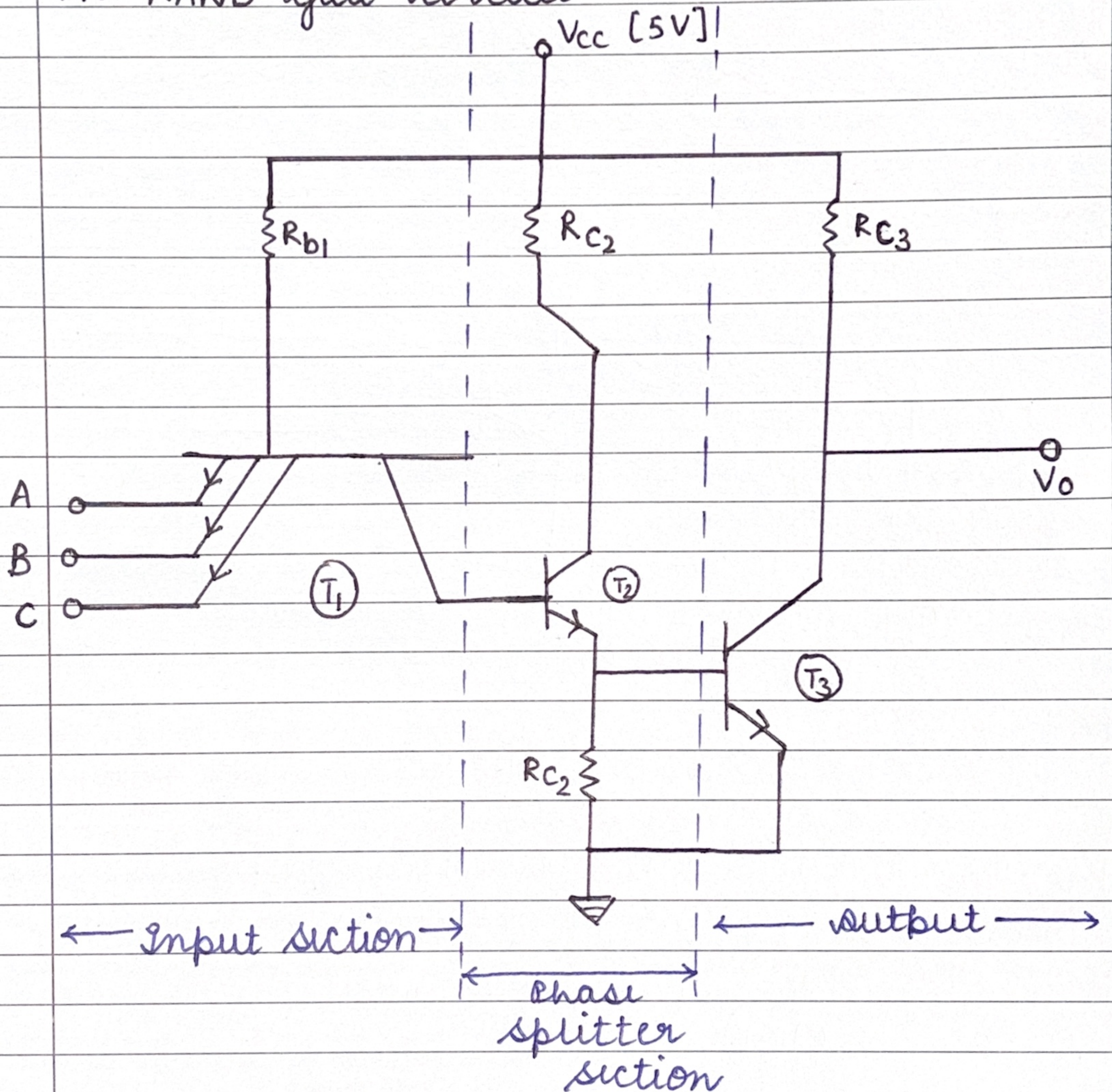
→ Transistor Transistor Logic (TTL)

1. Standard TTL
2. TTL totempole output
3. TTL open collector output
4. TTL tristate

} Classification of TTL

- TTL offers way higher speed as compared to DTL and RTL

★
→ TTL NAND gate circuit



A	B	C	V_0
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1

A	B	C	V_0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

→ ★

Working mechanism of TTL NAND gate

1. Case I → Atleast one output is low

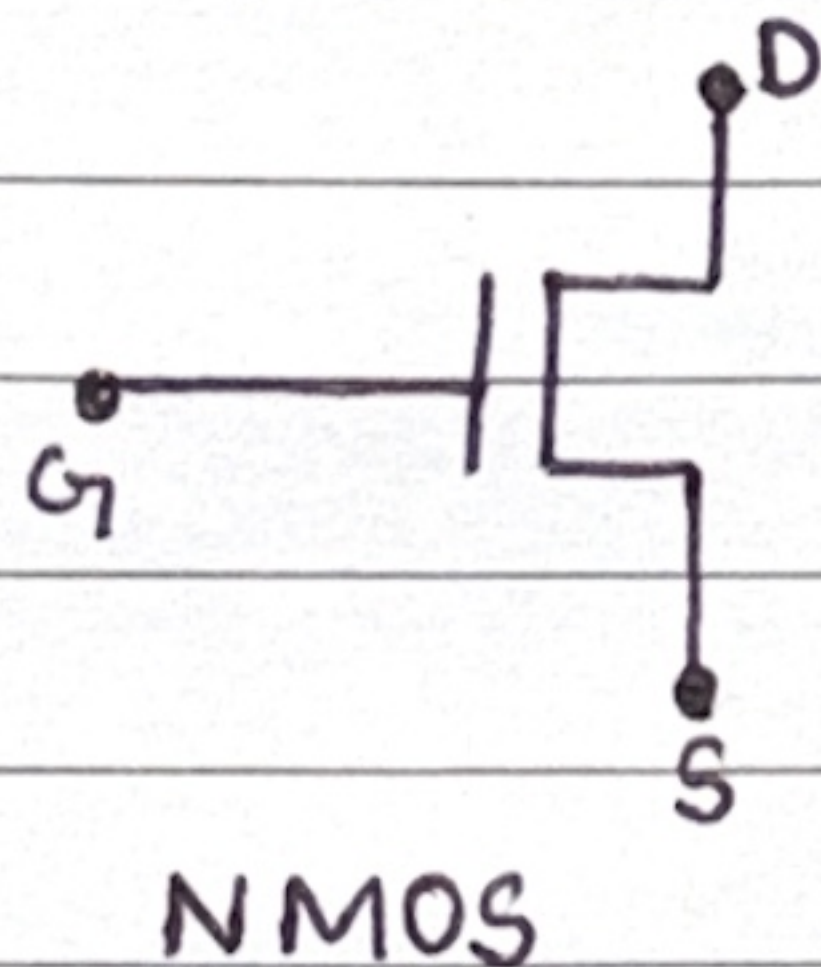
- Connecting any input to 0V forward biases the Base emitter junction of T_1
- Base current from V_{CC} flows out via T_1 's emitter to ground
- No base current reaches phase splitter so it remains off
- Since phase splitter is off, no current flows into T_2 's base so T_2 also remain off
- Phase splitter's collector voltage rises towards $+V_{CC}$ turning on T_3
- Output node connects to $+V_{CC}$ through T_4 & D_1
- Output V_O is high

2. Case II → Both inputs are high

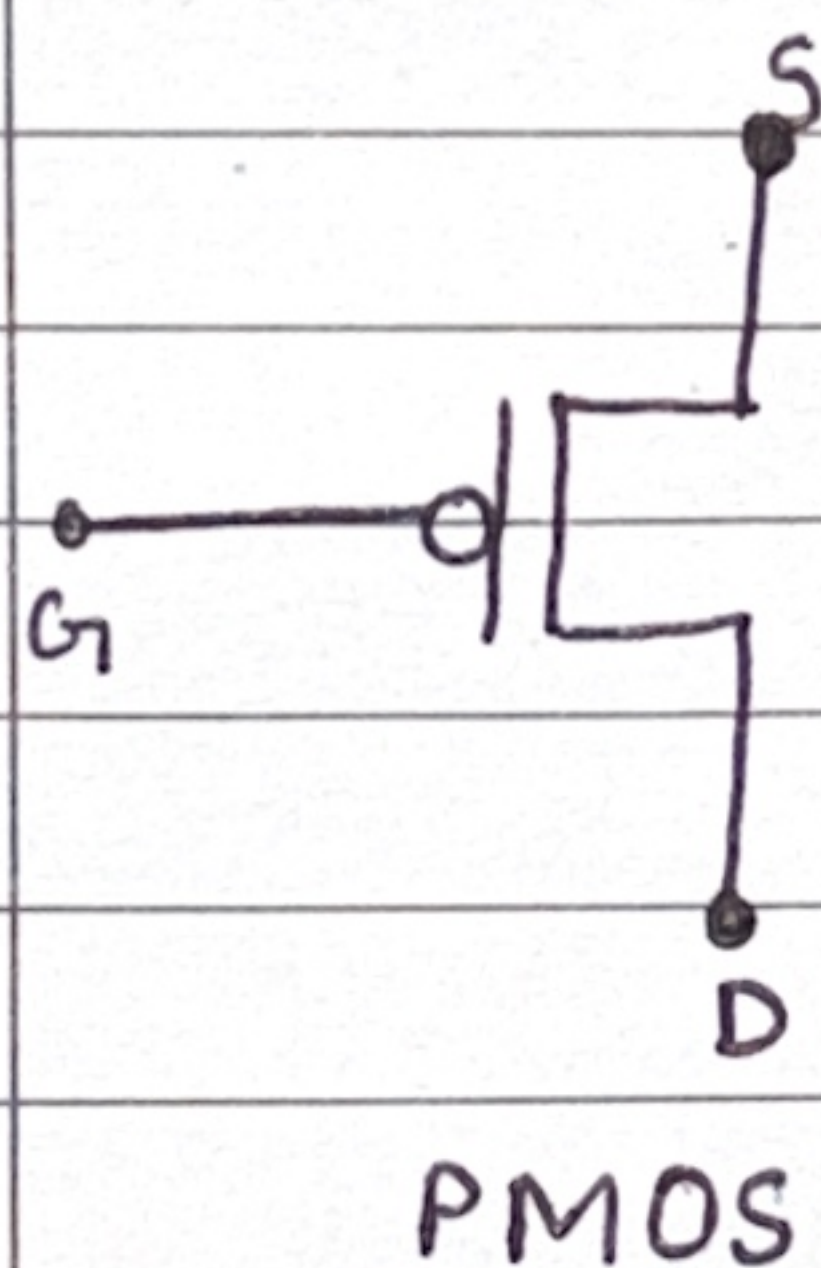
- Both high inputs reverse bias both Base emitter junctions of T_1
- Current from V_{CC} flows through T_1 's base collector junction directly into base of phase splitter
- Phase splitter is saturated & turns fully ON
- Voltage drop across resistor R_{C2} drives T_2 's base turning T_2 fully on.
- Phase splitter's collector voltage drops to $\sim 0.9V$ which is too low to turn on T_3 thus T_3 turns off
- Output V_O is low

→ CMOS

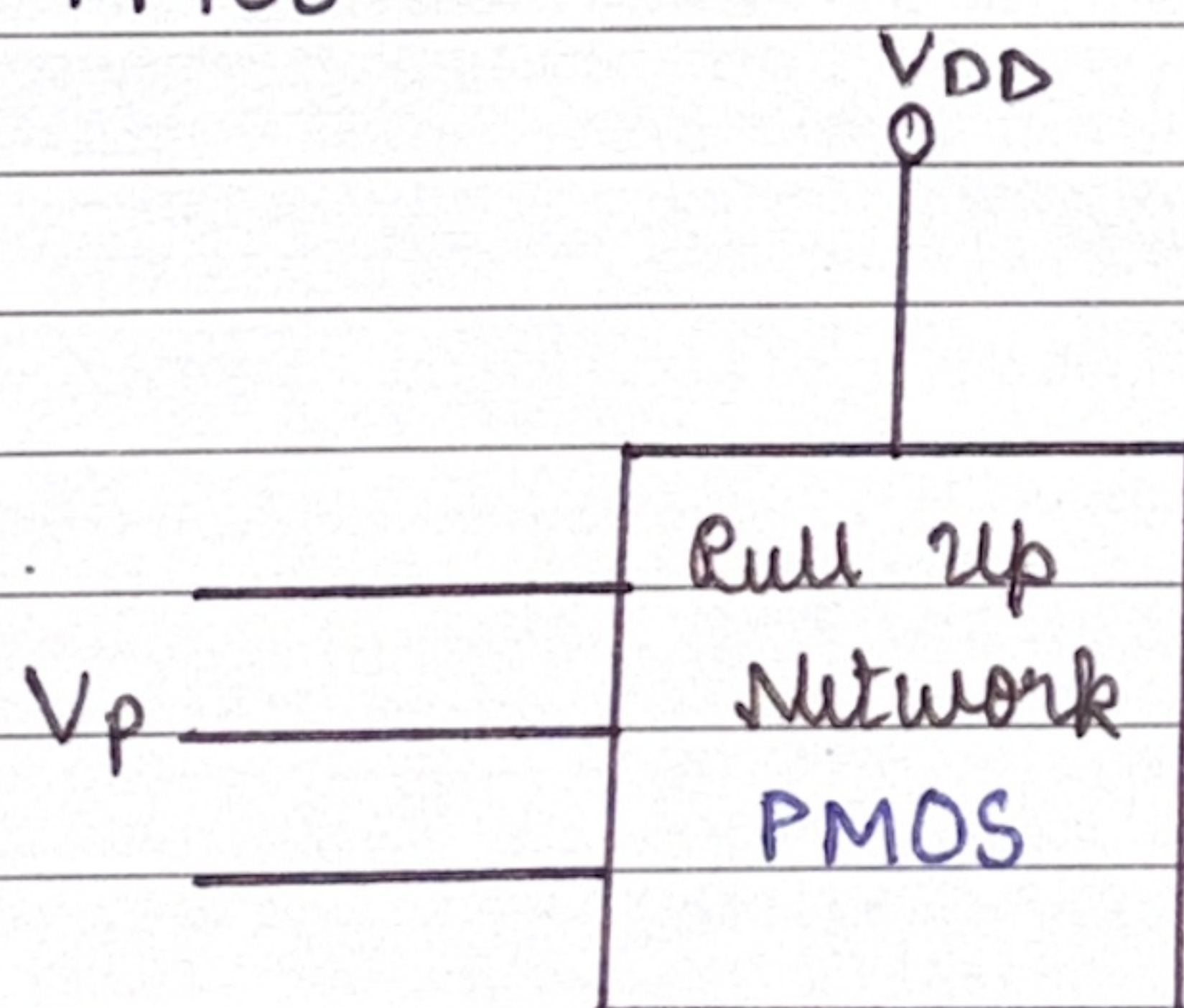
CMOS stands for complementary metal oxide semiconductor field effect transistor



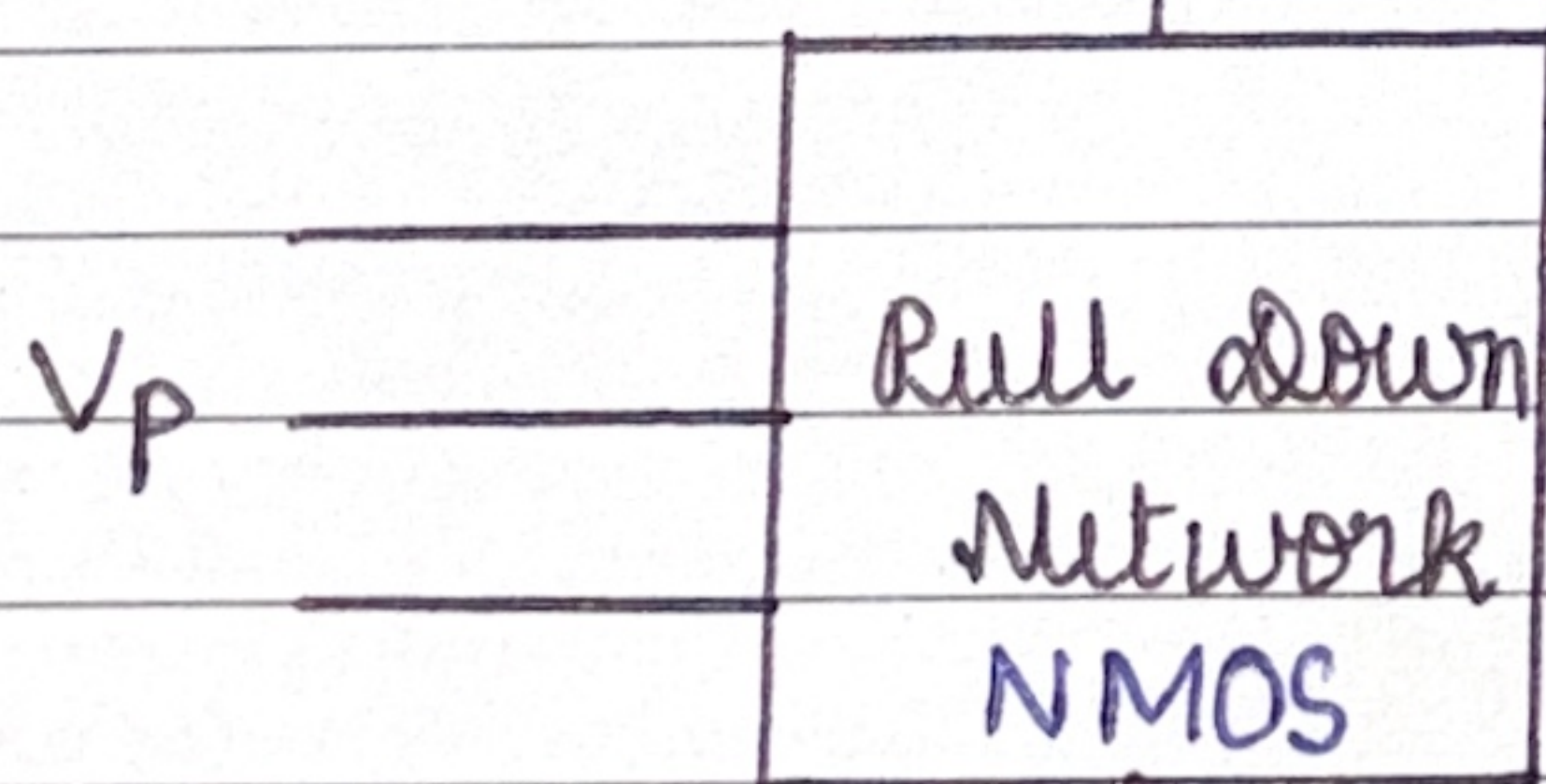
- If $G=0$, OFF, D to S acts as open circuit
- If $G=1$, ON, D to S acts as short circuit



- If $G=0$, ON, S to D acts as short circuit
- If $G=1$, OFF, S to D acts as open circuit



→ PMOS transistor can easily pass logic high



→ NMOS transistor can easily pass logic low

→ Rules for logic circuits using CMOS

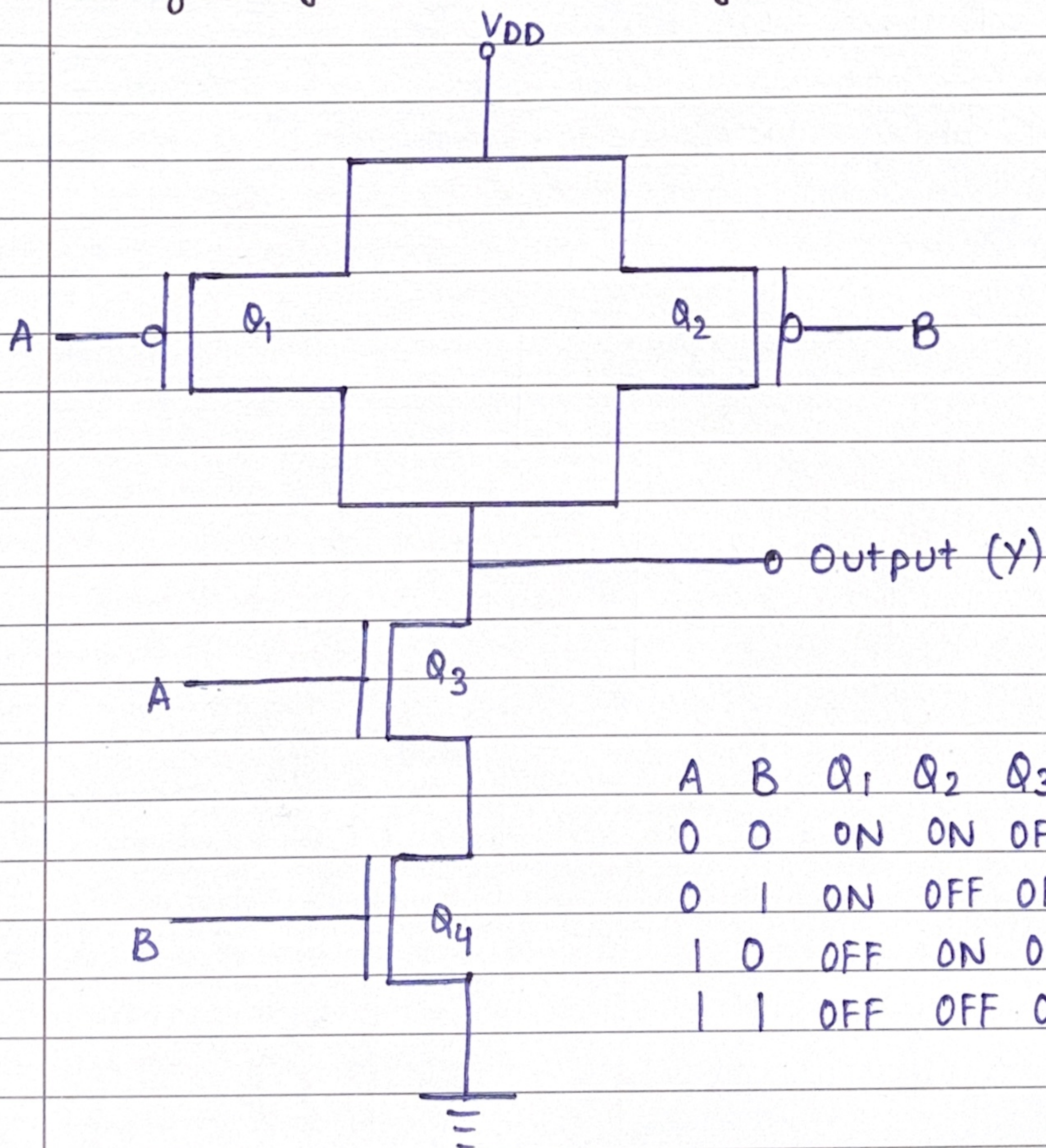
1. $(\cdot)$ operator [AND]

- PMOS → Parallel
- NMOS → Series

2. $(+)$ operator [OR]

- PMOS → Series
- NMOS → Parallel

→ Design of CMOS NAND gate



A	B	Q ₁	Q ₂	Q ₃	Q ₄	Y
0	0	ON	ON	OFF	OFF	1
0	1	ON	OFF	OFF	ON	1
1	0	OFF	ON	ON	OFF	1
1	1	OFF	OFF	ON	ON	0